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DUNMAN HIGH SCHOOL **Preliminary Examination** **Year 6**

H2 BIOLOGY

9648/02

Paper 2 Structured and Free-Response Questions

19 September 2016

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section B Structured Questions

Answer **all** questions.

Write your answers on space provided in the Question Paper.

Section C Free-Response Questions

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate. Write your answers on the writing paper provided.

Submit your answers to Sections B and Section C separately.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

For Examiner's Use	
Section A [40]	
Section B [80]	
1	/ 10
2	/12
3	/ 10
4	/ 12
5	/ 11
6	/ 9
7	/ 7
8	/ 9
Section C [20]	
1 / 2	
Total [140]	

This document consists of **22** printed pages.

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Section B: Structured Questions (80 marks)

Answer **all** questions in this section.

For
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Question 1

Triglycerides are not transported in the blood on their own as they are insoluble in water. Instead they are transported within lipoproteins such as LDL. Lipoproteins are made up of proteins and lipids. Their function is to carry cholesterol, triglycerides and other lipids through the blood. Lipoproteins such as LDL are then taken up by target cells via receptor mediated endocytosis.

Fig. 1.1 illustrates the structure of a LDL.

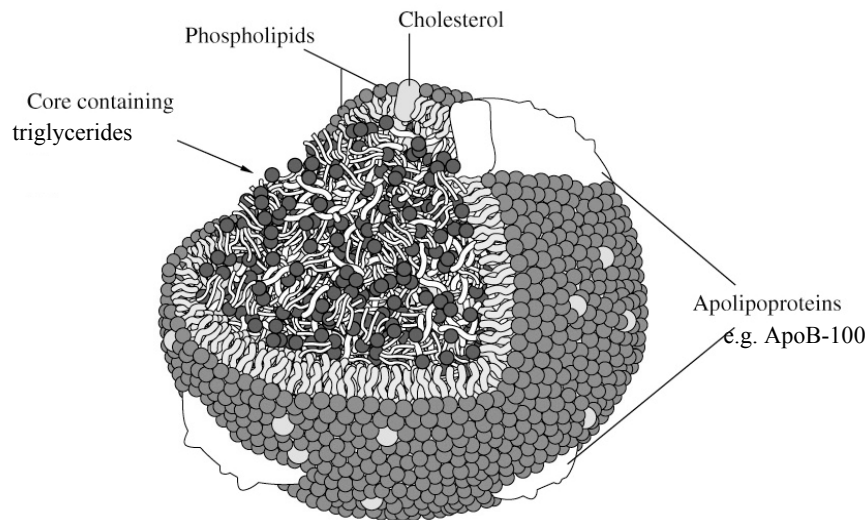


Fig. 1.1

Fig. 1.2 below shows the uptake of an LDL particle into a cell.

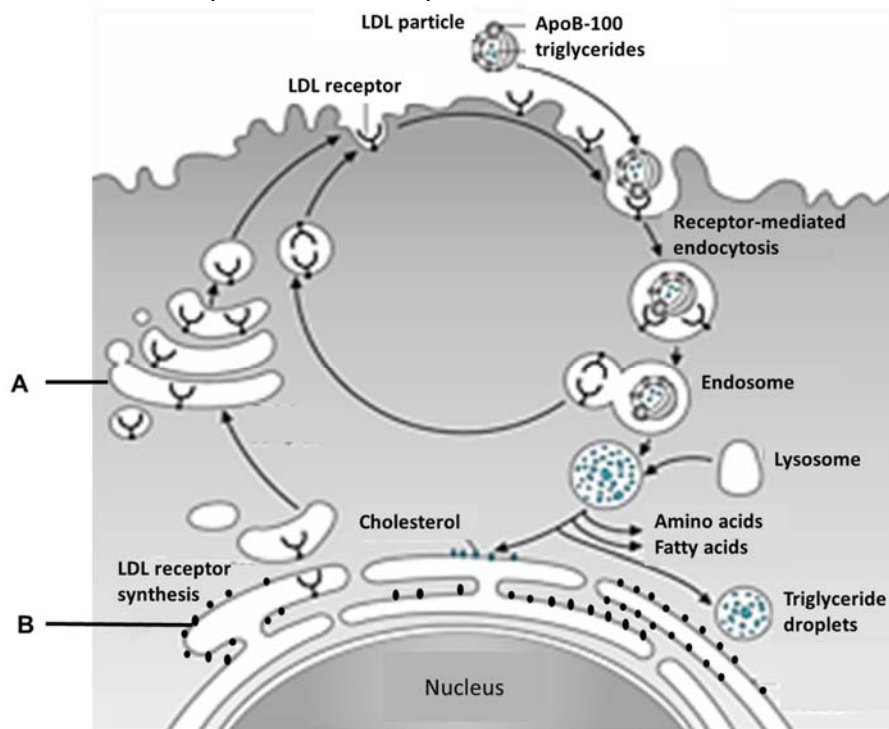


Fig. 1.2

- (a) With reference to **Fig. 1.1**, explain the role of phospholipids in lipoproteins such as LDL. [2]

- (b) With reference to **Fig 1.2**,

- (i) Name the organelles labelled **A** and **B**. [2]

A :

B :

- (ii) Explain the roles of organelles **A** and **B** in expression of the LDL receptor on the cell surface. [4]

A :

B :

- (c) Using the fluid mosaic model, explain how the properties of the cell surface membrane enable the uptake of LDL by a cell. [2]

Total:[10]

Question 2

- (a) **Fig. 2.1** shows the sequence of bases in a section of a single-stranded RNA virus. The bases code for the **first few amino acids** of a polypeptide chain.

5' UACAUGGAUUACCCCGUUGUACAU 3'

Fig. 2.1

Each codon codes for a specific amino acid as shown in **Table 2**.

Table 2

UUU	phe	UCU	ser	UAU	tyr	UGU	cys
UUC	phe	UCC	ser	UAC	tyr	UGC	cys
UUA	leu	UCA	ser	UAA	STOP	UGA	STOP
UUG	leu	UCG	ser	UAG	STOP	UGG	trp
CUU	leu	CCU	pro	CAU	his	CGU	arg
CUC	leu	CCC	pro	CAC	his	CGC	arg
CUA	leu	CCA	pro	CAA	gln	CGA	arg
CUG	leu	CCG	pro	CAG	gln	CGG	arg
AUU	ile	ACU	thr	AAU	asn	AGU	ser
AUC	ile	ACC	thr	AAC	asn	AGC	ser
AUA	ile	ACA	thr	AAA	lys	AGA	arg
AUG	met	ACG	thr	AAG	lys	AGG	arg
GUU	val	GCU	ala	GAU	asp	GGU	gly
GUC	val	GCC	ala	GAC	asp	GGC	gly
GUA	val	GCA	ala	GAA	glu	GGA	gly
GUG	val	GCG	ala	GAG	glu	GGG	gly

Using information from **Fig. 2.1** and **Table 2**,

- (i) State the **third** amino acid coded by the section shown in **Fig. 2.1** if the virus was a **positive-sense** RNA virus. [1]

- (ii) State the **fourth** amino acid coded by the section shown in **Fig. 2.1** if the virus was a **negative-sense** RNA virus. [1]

- (b) (i)** Termination of protein synthesis is not 100% efficient. A number of natural mechanisms that suppress translation termination exist. One of them is the 'STOP codon readthrough'. This process enables the ribosome to pass through the STOP codon in mRNA and continue translation to the next STOP codon. STOP codon readthrough is commonly observed in viruses.

Suggest an advantage of STOP codon readthrough for viruses. [1]

- (ii)** 'Nonstop' mutations are single base-pair substitutions that occur within translational termination (stop) codons.

State the immediate events that would occur when the ribosome reaches one such 'nonstop' mutation during translation of an mRNA in yeast. [2]

- (iii)** Contrast 'nonstop' mutation with 'nonsense' mutation. [1]

- (c) Cystic fibrosis is a genetic disorder that affects the respiratory and digestive systems. People with cystic fibrosis inherit a defective gene on chromosome 7 called CFTR (cystic fibrosis transmembrane conductance regulator) gene.

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Well over one thousand mutations have been described that can affect the CFTR gene. Two of such mutations are shown in **Fig. 2.2**.

Amino acid position	506	507	508	509	510		522	523	524
Normal CFTR gene	ATC	ATC	TTT	GGT	GTT		GCA	TGC	CAA
Mutation 1	ATC	ATT	GGT	GTT	GCC		GCA	TGC	CAA
Mutation 2	ATC	ATC	TTT	GGT	GTT		GCA	TGA	CAA

Fig. 2.2 (showing part of the base sequence on the non-template DNA strand)

Table 2

UUU	phe	UCU	ser	UAU	tyr	UGU	cys
UUC	phe	UCC	ser	UAC	tyr	UGC	cys
UUA	leu	UCA	ser	UAA	STOP	UGA	STOP
UUG	leu	UCG	ser	UAG	STOP	UGG	trp
CUU	leu	CCU	pro	CAU	his	CGU	arg
CUC	leu	CCC	pro	CAC	his	CGC	arg
CUA	leu	CCA	pro	CAA	gln	CGA	arg
CUG	leu	CCG	pro	CAG	gln	CGG	arg
AUU	ile	ACU	thr	AAU	asn	AGU	ser
AUC	ile	ACC	thr	AAC	asn	AGC	ser
AUA	ile	ACA	thr	AAA	lys	AGA	arg
AUG	met	ACG	thr	AAG	lys	AGG	arg
GUU	val	GCU	ala	GAU	asp	GGU	gly
GUC	val	GCC	ala	GAC	asp	GGC	gly
GUA	val	GCA	ala	GAA	glu	GGA	gly
GUG	val	GCG	ala	GAG	glu	GGG	gly

Using information from **Fig. 2.2** and **Table 2**,

- (i) Explain the effect of mutation 1 on the structure of the protein formed. [4]

(ii) Explain the effect of mutation 2 on the amino acid sequence in the protein. [2]

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Total: [12]

Question 3

Fig. 3.1 shows the first step in T4 infection of its host.

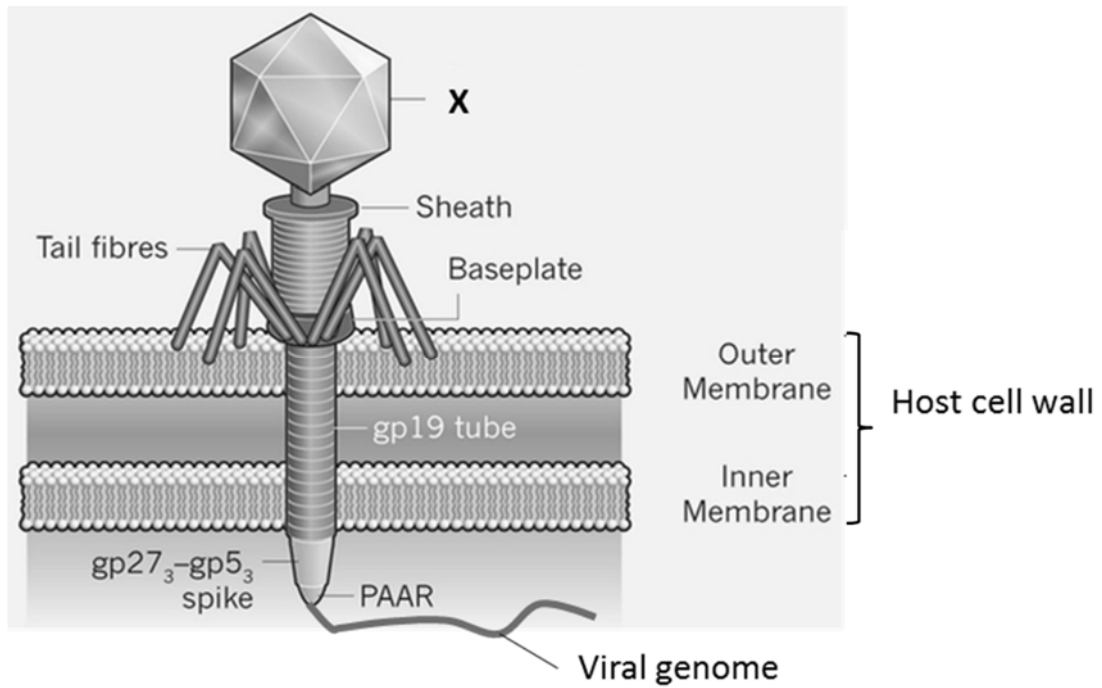


Fig. 3.1

- (a)** Name structure **X** and explain its function. [2]

- (b)** Explain the role of the contractile sheath and as shown in **Fig. 3.1**. [2]

- (c)** Describe what occurs after the stage shown in **Fig. 3.1** to complete the virus life cycle. [4]

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- (d)** Explain how the T4 phage can result in horizontal gene transfer between bacteria. [2]

Total: [10]

Question 4

- (a) **Fig. 4.1** shows the different phases of the cell cycle. The arrows indicate the checkpoints of the cell cycle.

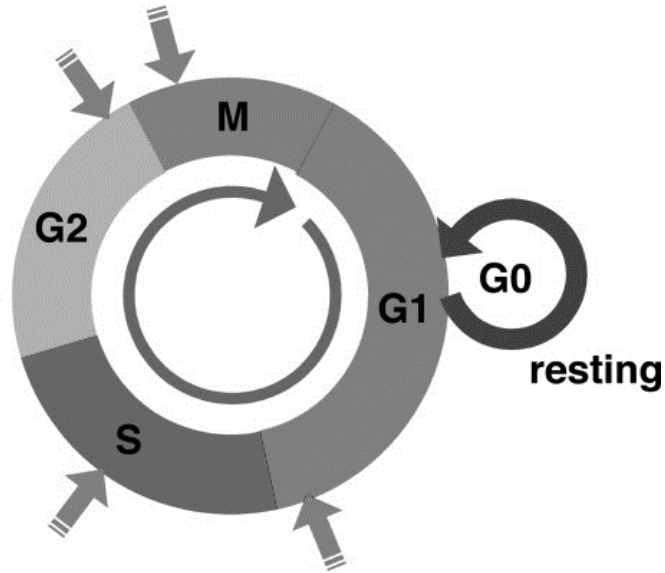


Fig. 4.1

- (i) Outline how the normal mitotic cell cycle is regulated at the G_1 and M checkpoints. [4]

- (ii) G_2 is part of a stage that takes place during the cell cycle. Describe what happens during this stage. [2]

- (iii) Upon maturity, nerve and heart muscle cells enter into a G_0 phase that can last indefinitely. Such cells are said to be quiescent. On the other hand, cells such as fibroblasts can reach a maximum of 50 cell divisions before becoming senescent.

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State two differences between cellular quiescence and senescence. [2]

- (b) Fig. 4.2 below shows a cell at a certain stage of nuclear division in *Drosophila*.

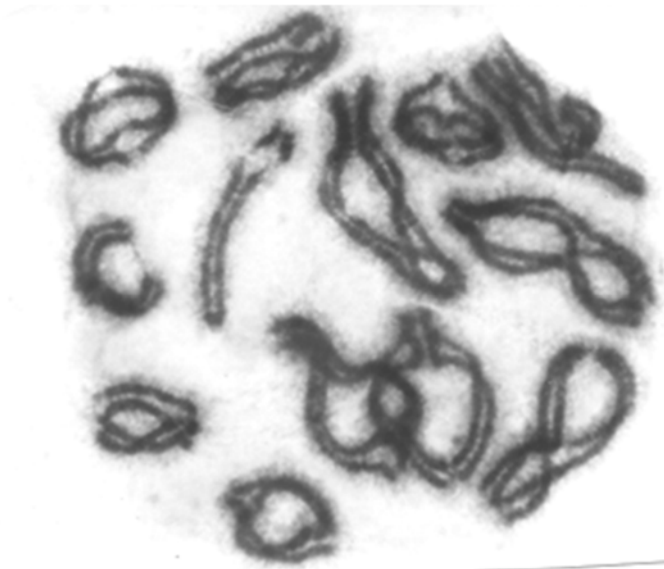


Fig. 4.2

- (i) State the number of telomeres present in the cell. Explain your answer. [2]

- (ii) Explain the role centromeres play in mitosis. [2]

Total: [12]

Question 5*For
Examiner's
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- (a) Malvidin is a plant pigment responsible for the colours of red grapes, cranberries and blueberries which may have anticancer properties. The dominant allele, K, codes for an enzyme involved in the biosynthesis of malvidin. The presence of dominant allele, D, of another unlinked gene, results in the absence of malvidin production in plants, even when the enzyme is present whilst the recessive allele, d, does not affect malvidin production.
- (i) Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the F1 and F2 generations of a cross between a pure-breeding malvidin-producing plant and a non-producing plant of genotype k'k'DD. Give the ratio of the phenotypes in the resulting F2 generation. [4]

(ii) Explain how the two genes interact. [2]

For
Examiner's
use

- (b) The dominant allele, S, of another gene results in the fruits having smooth skin, whilst the recessive allele, s', results in the fruits having wrinkled skin. A cross was made between a malvidin-producing plant with smooth-skinned fruits (KKSS) and a non-producing plant with wrinkled-skinned fruits (k'k's's'). The F₁ generation were all malvidin-producing plants with smooth-skinned fruits. The F₁ plants were test crossed and gave offspring with the following numbers of plants in each of the four phenotypes:

Malvidin-producing plants with smooth-skinned fruits	40
Non-producing plants with wrinkled-skinned fruits	42
Malvidin-producing plants with wrinkled-skinned fruits	20
Non-producing plants with smooth-skinned fruits	18

Distribution of χ^2

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 \text{ test: } \chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Key to symbols: Σ = sum of ...
 v = degrees of freedom
 c = number of classes
 O = observed value
 E = expected value

- (i) Calculate the χ^2 value for the given data. Show your working below. [2]

For
Examiner's
use

- (ii) Using the information from the table provided, explain the conclusion drawn from the calculated χ^2 value above. [3]

Total:[11]

Question 6

- (a) The Korean rice beer *takju* is prepared by mixing a yeast fermentation starter powder (*nuruk*) with cooked rice and incubating at approximately 20°C for 2-3 days, following which it is filtered through a fine mesh. **Fig. 6.1** shows the summary of this process.

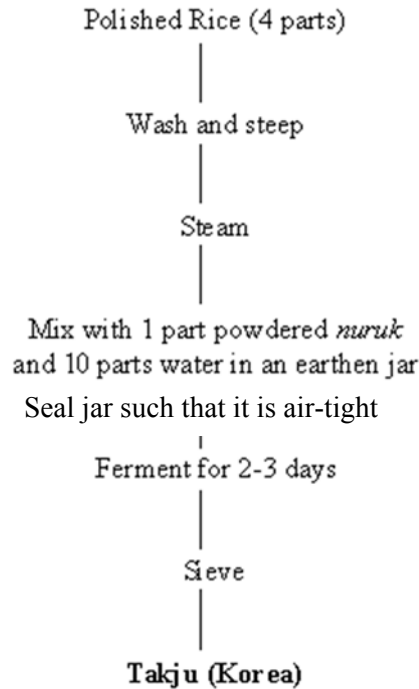


Fig. 6.1

Biochemical changes occurring during the fermentation of *takju* are summarised in **Fig. 6.2**.

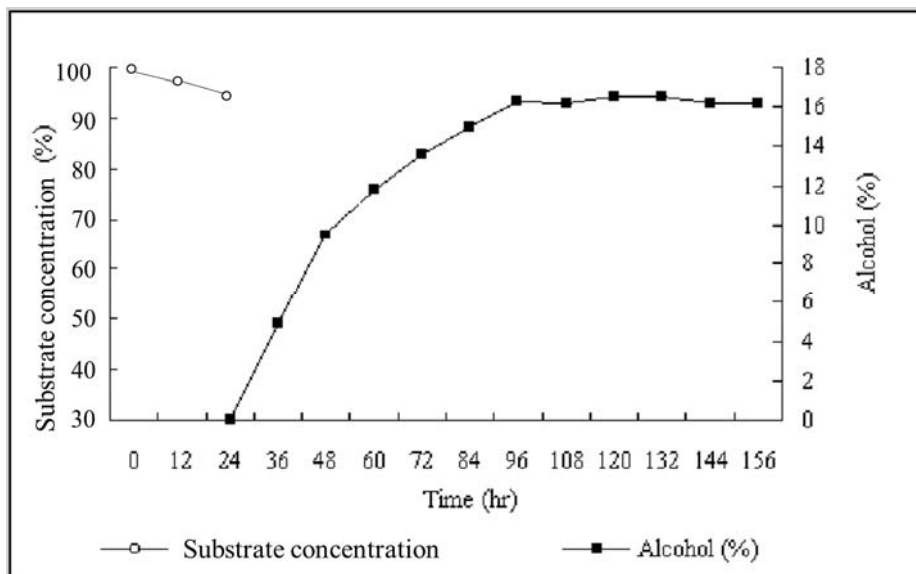


Fig. 6.2

(a) With reference to **Fig. 6.1** and **Fig. 6.2**,

(i) Explain why there was a lag time of 24h before any alcohol was produced. [1]

(ii) Explain the mode of respiration of the yeast between 24-96h and relate it to the production of alcohol in the brewing of *takju*. [5]

(iii) Explain why there is no further increase of alcohol concentration after 96h. [2]

(b) **Fig. 6.2** shows the changes in substrate concentration in the brewing of *takju* from 0-24h.

Complete the graph on **Fig. 6.2** to show the predicted changes in substrate concentration from 24-96h. [1]

Total: [9]

Question 7

- (a) GABA is a neurotransmitter present in some parts of the nervous system. **Fig. 7.1** shows how the release of GABA from a presynaptic membrane affects the ion channels of a postsynaptic membrane.

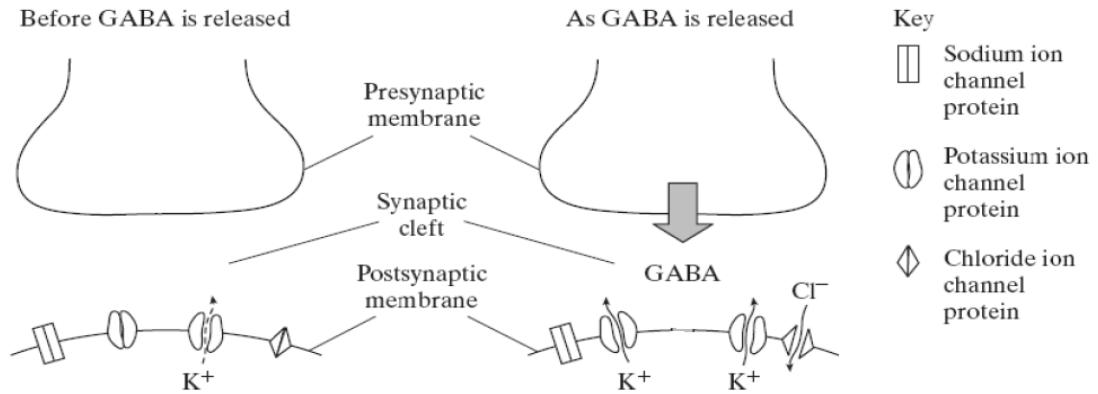


Fig. 7.1

- (i) With reference to **Fig. 7.1**, explain what would happen to the membrane potential on the postsynaptic membrane when GABA is released. [2]

Fig. 7.2 shows the synapses of neurone **B** with two other neurones, **P** and **Q**.

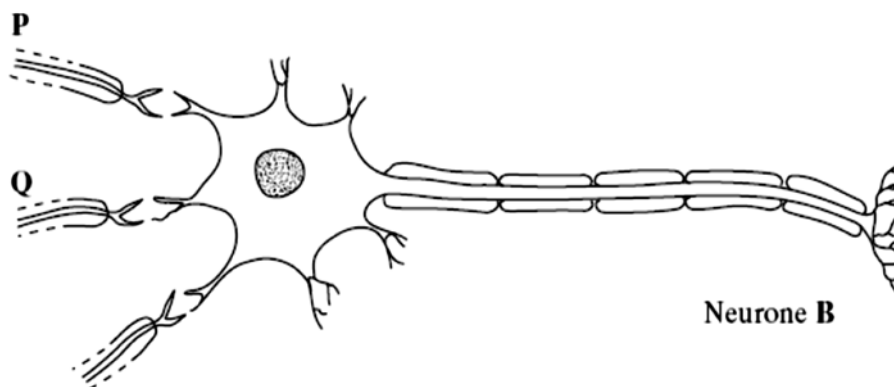


Fig. 7.2

Neurone **P** releases acetylcholine whereas neurone **Q** releases GABA.

- (ii) Explain why neurone **B** is less likely to respond if both neurones **P** and **Q** are stimulated at the same time. [2]

*For
Examiner's
use*

- (iii) Barbiturates act as depressants with effects similar to anesthetics. They act mainly by enhancing the activity of the GABA neurotransmitter. Suggest how barbiturates enhance the activity of the GABA neurotransmitter. [1]

- (b) Explain how the temporal summation of various stimuli can result in a coordinated response at the postsynaptic neurone through post synaptic potentials. [2]

Total:[7]

Question 8

- (a) Fishes in the genus *Anisotremus* comprise of ten described species which occur predominantly on coral reefs and subtropical rocky reefs in the Neotropics.

Bernardi *et.al.* did a molecular phylogenetic study on such fishes in that area. Results are shown in **Fig. 8.1**.

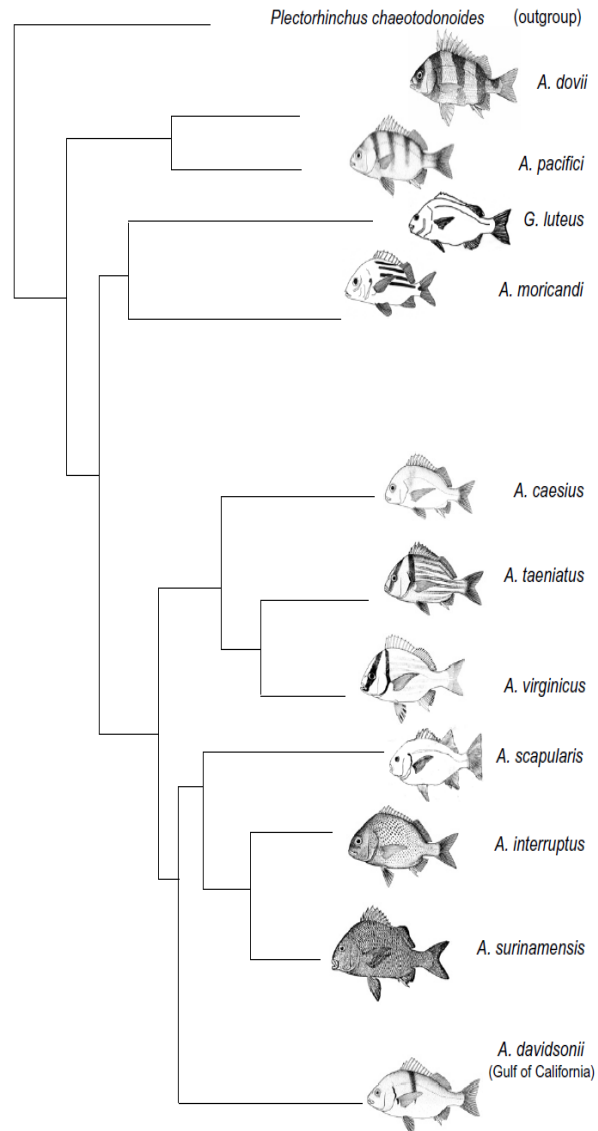


Fig. 8.1

- (i) Explain how molecular methods can be used to elucidate the evolutionary relationships of the different species of *Anisotremus* fishes. [2]

- (ii) Explain why it may be more reliable to construct a phylogenetic tree of the ten species of *Anisotremus* using molecular data instead of morphological comparisons. [2]

For
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- (iii) With reference to **Fig. 8.1** and molecular homology, comment on the evolutionary relationship between *A. taeniatus* and *A. virginicus*. [1]

- (b) *A. taeniatus* is found in the Pacific Ocean, whereas *A. virginicus* is found in the Caribbean Sea. These two species were derived due to the formation of the Isthmus of Panama about 3.5 million years ago. Before that event, the waters of the Pacific Ocean and Caribbean Sea mixed freely.

For
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use

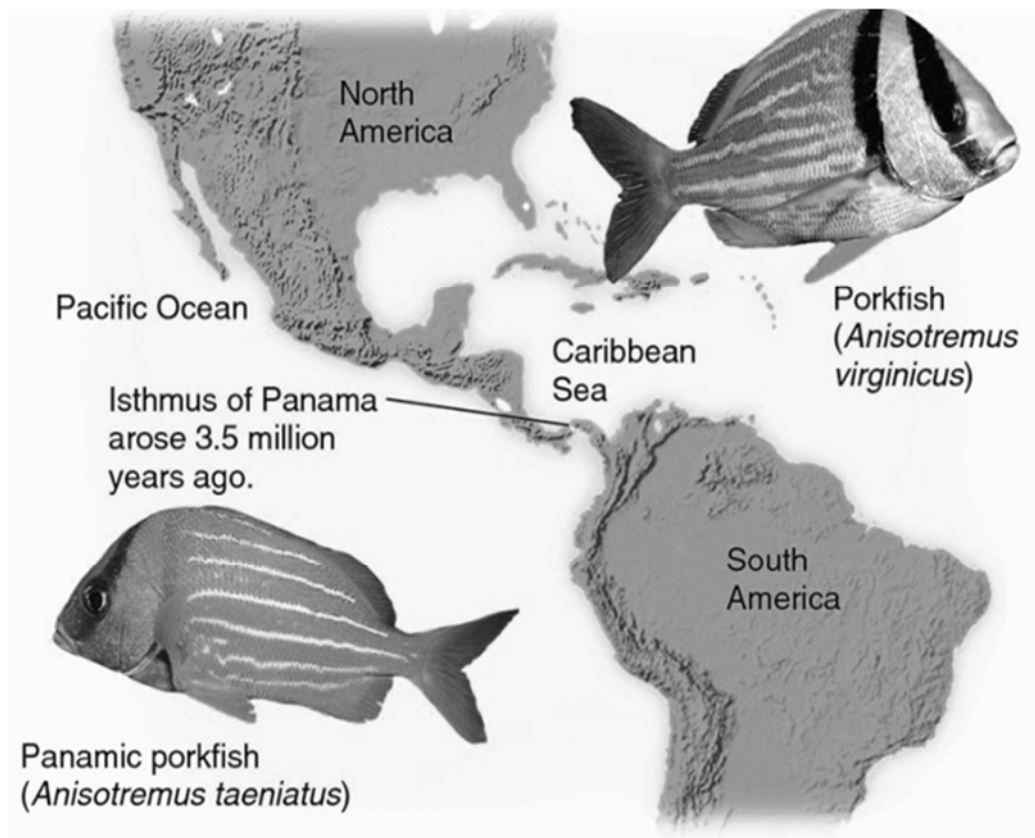


Fig. 8.2

Explain how the formation of the Isthmus of Panama results in the emergence of *A. taeniatus* and *A. virginicus*. [4]

Total:[9]

Section C: Free-Response Question (20 marks)

Answer only **one** question.

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 1

- (a) Describe the protein folding of an enzyme and relate its structure to its function. [10]
- (b) Describe the effect of pH on enzymes and their activity. [4]
- (c) Compare and contrast competitive and non-competitive inhibitors and their effects on the rate of enzyme activity. [6]

OR

Question 2

- (a) Distinguish between the processes of Krebs Cycle and Calvin Cycle. [8]
- (b) State the similarities between ATP production in mitochondria and chloroplasts and suggest why these similarities exist. [6]
- (c) Discuss the effects of varying carbon dioxide and oxygen levels on photosynthesis. [6]

Total: [20]

END OF PAPER